

1	ACSBR/2025/I/03
	Solve the equation $4(3^x) + 5(3^{1-x}) = 3^{x+2} + 10$. [5]
2	ACSBR/2025/I/07
	In an experimental environment, the population of a type of insect was observed. After t days, the number of insects N was given by the equation $N = 500 + 5000e^{kt}$, where k is a constant. (a) Find the initial population of the insects. [1] (b) Given that the population of the insects decreased to 5000 after 3 days, find the value of k correct to 3 significant figures. [3] (c) After a long period of time, the population of the insects approaches a particular value A. Find the value of A. [2]
3	ACSBR/2025/I/02
	Express $2\log_3 x - \log_3(x-5) = 2$ as a quadratic equation in x and explain why there are no real solutions. [5]
4	BASS/2025/I/12
	(a) Given that $(\log_x y)^2 + 27\log_y x = 0$ where $x > 0$ and $y > 1$, express y as a power of x. [4] (b) Solve the equation $2\log_4(x+1) + \log_2(2x-5) = 2$. [6] Explain why one of the answers is rejected.
5	BASS/2025/II/04
	The mass, m grams, of a radioactive substance, present t hours after first being observed, is given by the formula $m = m_0 e^{-kt}$ where m_0 and k are positive constants. The mass of the substance has a mass of 150 grams initially. After 6 hours, its mass has decreased to 110 grams. (a) Find the value of m_0 and k. [3] (b) Calculate the time taken, to the nearest hour, for the substance to decrease to half of its original mass. [2] (c) Explain what the mass of the radioactive substance is after a long period of time. [1]
6	CCHY/2025/II/1
	In 2020, there were an estimated 7 million insects of a particular species in a certain country. In 2021, the numbers were estimated to have fallen to 5 million. Scientists believe that the number of these insects, N million, can be modelled by the formula $N = 3 + ae^{-kt}$, where t is the time in years after 2020. (a) Find the exact values of a and of k. [4] (b) What is the approximate size of the population for large values of t? [1]

7	CCHY/2025/II/6
	(a) Solve the equation $\log_5(3y+1) - \frac{1}{\log_y 5} + \log_2 8 = 4$. [4] (b) By expressing the equation $\ln(6e^x + 1) + x = 0$ as a quadratic equation in e^x, solve the equation $\ln(6e^x + 1) + x = 0$, giving values of x in logarithmic form. [4]
8	GMS/2025/II/1
	A man buys a new vehicle. After t months its value, $\\$V$, is given by $V = 10000 e^{-pt}$, where p is a constant. (a) Find the initial value of the vehicle. [1] The value of the vehicle after 12 months is expected to be \$4000. Calculate (b) the expected value of the vehicle after 18 months, [3] (c) the age of the vehicle, to the nearest month, when its expected value will be \$1000. [2]
9	GMS/2025/II/7
	Solve the following equations. (a) $\log_2 x + \frac{2}{\log_2 x} = 3$ [4] (b) $\log_3(x+1) - \log_{\sqrt{3}}(x-1) = 1$ [6]

1	$4\left(3^x\right)+5\left(\frac{3}{3^x}\right)=3^x \cdot 9+10$ $4y+\frac{15}{y}=9y+10$ $4y^2+15=9y^2+10y$ $5y^2+10y-15=0$ $y^2+2y-3=0$ $(y+3)(y-1)=0$ $y=-3$ $3^x=-3 \text{ (reject, } 3^x>0)$ $y=1$ $3^x=3^0$ $x=0$
2	(a) $N=500+5000e^0$ $=5500$ $5000e^{3k}+500=5000$ $e^{3k}=0.9$ (b) $k=\frac{1}{3}\ln 0.9$ $=-0.0351 \text{ (to 3 sig fig)}$ (c) $\text{As } t \rightarrow \infty, 5000e^{-kt} \rightarrow 0, N \rightarrow 500$ $A=500$
3	$\log_3 x^2-\log_3(x-5)=2$ $\log_3 \frac{x^2}{x-5}=2$ $\frac{x^2}{x-5}=3^2$ $\frac{x^2}{x-5}=9$ $x^2-9x-45=0$ $b^2-4ac=(-9)^2-4(-9)(-45)$ $=-99<0$ Since the discriminant is less than 0, there are no real solutions.

4	a	$(\log_x y)^2 + 27 \log_y x = 0$ $(\log_x y)^2 + \frac{27}{\log_x y} = 0$ Let $\log_x y$ be u $u^2 + \frac{27}{u} = 0$ $\frac{u^2 + 27}{u} = 0$ $u^3 = -27$ $u = -3$ $\log_x y = -3$ $y = x^{-3}$
	b	$2 \log_4(x+1) + \log_2(2x-5) = 2$ $\frac{2 \log_2(x+1)}{\log_2 4} + \log_2(2x-5) = 2$ $\frac{2 \log_2(x+1)}{2 \log_2 2} + \log_2(2x-5) = 2$ $\log_2(x+1)(2x-5) = 2$ $(x+1)(2x-5) = 4$ $2x^2 - 3x - 5 = 4$ $2x^2 - 3x - 9 = 0$ $(2x+3)(x-3) = 0$ $x = -\frac{3}{2} \text{ (rejected) or } x = 3$ When $x = -\frac{3}{2}$, $\log_4(x+1) = \log_4\left(\frac{-1}{2}\right)$ $\log_2(2x-5) = \log_2(-8)$ are undefined.
5	a	$t = 0,$ $150 = m_o e^0$ $m_o = 150$

	$110 = 150e^{-k(6)}$ $e^{-6k} = \frac{110}{150}$ $-6k = \ln\left(\frac{11}{15}\right)$ $k = 0.051692 \text{ (5sf)}$ $k = 0.0517 \text{ (3sf)}$
	b $75 = 150e^{-0.051692t}$ $e^{-0.051692t} = 0.5$ $-0.051692t = \ln 0.5$ $t = 13.409 \text{ (5 sf)}$ $t = 14 \text{ (nearest hour)}$
	c $\because t \rightarrow \infty$ $\therefore e^{-0.051692t} \rightarrow 0$ $\therefore 150e^{-0.051692t} \rightarrow 0$ $\therefore m \rightarrow 0$
6	(a) When $t = 0$, $N = 7$. $7 = 3 + ae^{-k(0)}$ $a = 7 - 3$ $= 4$ When $t = 1$, $N = 5$. $5 = 3 + 4e^{-k(1)}$ $2 = 4e^{-k}$ $e^{-k} = \frac{2}{4}$ $-k = \ln \frac{1}{2}$ $k = \ln 2$ (b) $N = 3 + 4e^{-(\ln 2)t}$ When t is large, $e^{-(\ln 2)t}$ becomes close to zero. Thus, the approximate size is 3 million.

7

(a)

$$\log_5(3y+1) - \frac{1}{\log_y 5} + \log_2 8 = 4$$

$$\log_5(3y+1) - \log_5 y = 4 - \log_2 8$$

$$\log_5\left(\frac{3y+1}{y}\right) = 4 - \log_2 2^3$$

$$\log_5\left(3 + \frac{1}{y}\right) = 4 - 3$$

$$3 + \frac{1}{y} = 5^1$$

$$\frac{1}{y} = 2$$

$$y = \frac{1}{2}$$

(b)

$$\ln(6e^x + 1) + x = 0$$

$$\ln(6e^x + 1) = -x$$

$$6e^x + 1 = e^{-x}$$

$$6(e^x)^2 + (e^x) - 1 = 0$$

$$(3e^x - 1)(2e^x + 1) = 0$$

$$e^x = \frac{1}{3} \quad \text{or} \quad e^x = -\frac{1}{2} \text{ (rej)}$$

$$x = \ln \frac{1}{3} \text{ or } -\ln 3$$

8

(a)

$$\text{When } t = 0, \quad V = 10000$$

(b)

$$\text{When } t = 12, \quad V = 4000$$

$$4000 = 10000 e^{-12p}$$

$$e^{-12p} = 0.4$$

$$-12p = \ln 0.4$$

$$p = \frac{\ln 0.4}{-12}$$

$$= 0.076357561$$

$$\text{When } t = 18, \quad V = 10000e^{-0.076357561(18)}$$

$$= 2529.80 \text{ (or 2530)}$$

(c)

$$\begin{aligned}\text{When } V = 1000, \quad 1000 &= 10000 e^{-0.076357561(t)} \\ e^{-0.076357561(t)} &= 0.1 \\ -0.076357561(t) &= \ln 0.1 \\ t &= 30.155 \\ &= 30\end{aligned}$$

9

(a)

$$\log_2 x + \frac{2}{\log_2 x} = 3$$

$$\text{Let } u = \log_2 x$$

$$u + \frac{2}{u} = 3$$

$$u^2 - 3u + 2 = 0$$

$$(u-1)(u-2) = 0$$

$$\begin{aligned}u = 1 & \quad \text{or} \quad u = 2 \\ \log_2 x = 1 & \quad \text{or} \quad \log_2 x = 2 \\ x = 2 & \quad \text{or} \quad x = 4\end{aligned}$$

(b)

$$\log_3(x+1) - \log_{\sqrt{3}}(x-1) = 1$$

$$\log_3(x+1) - \frac{\log_3(x-1)}{\log_3 \sqrt{3}} = 1$$

$$\log_3(x+1) - \frac{\log_3(x-1)}{\frac{1}{2}} = 1$$

$$\log_3(x+1) - 2\log_3(x-1) = 1$$

$$\log_3 \frac{(x+1)}{(x-1)^2} = 1 \quad [$$

$$\frac{(x+1)}{(x-1)^2} = 3$$

$$x+1 = 3(x-1)^2$$

$$x+1 = 3(x^2 - 2x + 1)$$

$$3x^2 - 7x + 2 = 0$$

$$(3x-1)(x-2) = 0$$

$$x = \frac{1}{3} \quad (\text{NA}) \quad \text{or} \quad x = 2$$